Limits of Landauer and second law

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1. Introduction

This document is a supplement to the reference paper "Limits of Landauer principle and of the second law of thermodynamics" [1] [2]. It explains the simulation scripts designed to demonstrate the article's conclusions.
We consider two types of memory inspired by the article by Jun et al.[3], which implement a particle evolving on a potential profile U(x,t) dependent on time t and abscissa x in a nanosystem.
- The device of Jun et al. is a bistable memory whose profile is tilted to force the particle to change state. We call it *tilting bistable memory*.
- A memory with a single potential well whose change of state is caused by a translation of the well. We call it *single well memory*.

2. Potential energy landscape

The first scheme of the experiments is a double well potential landscape to realize a one-bit memory. A particle is represented by its position x and its potential energy. The phase space of the system is shown by a continuous curve $U(x)$.
Our energy unit is $k_B T$, the entropy unit k_B, so the Clausius entropy inequation can be written $dS \geq dQ$.
Figure 1 shows a bistable memory with a symetric double well in state 1. The line separating the two states is conventionnaly at $x = 0$.

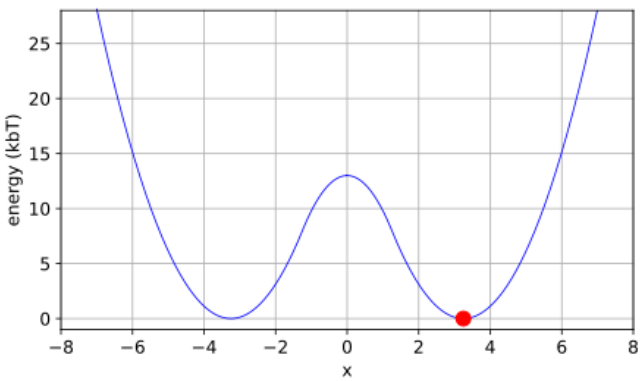

Figure 1 – Bistable memory in state 1

The following figure shows the energy transfers due to 3 causes : brownian mouvement, potential gradient $\frac{\partial}{dx}U(x,t)$ and action of actuator $\frac{\partial}{dt}U(x,t)$ changing the potential landscape.

The finite difference equations are detailed in the reference paper. They allow to simulate the evolution of the memory from an initial state $x0$ and $U(x0,t0)$. Due to the high stochasticity of Brownian motion, the variables must be averaged over a large number of trials.

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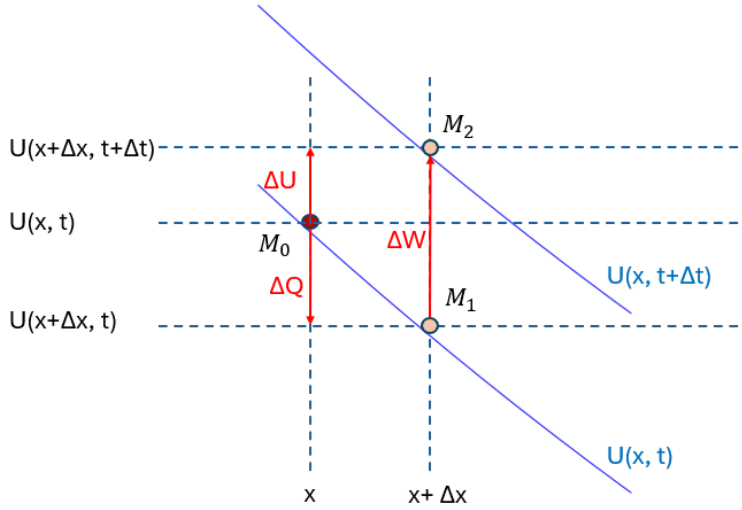

Figure 2 - Energy transfers during a period Δt

At the beginning of one period Δt, the particle is in M_0. During Δt, the particle moves by Δx from M_0 to M_1, due to potential gradient and to Brownian motion. An energy ΔQ is exchanged between the thermostat and the particle. At the end of the period, the energy landscape is suddenly modified from $U(x+\Delta x, t)$ to $U(x+\Delta x, t+\Delta t)$, exchanging energy ΔW between the actuator and the particle. The final balance is $\Delta U = \Delta Q + \Delta W$.

3. Tilting bistable memory

Jun et al. [3] made experiments to confirm Landauer's limit, which yielded highly accurate results. The energy landscape $U(x,t)$ is a quartic function of x:

$U(x,t) = Eb\,(x^4 - 2\,g(t)\,x^2 - 2\,f(t)x)$ with $Eb = 13$ (in $k_B\,T$ units)

phase	time interval	f(t)	g(t)
1 lowering barrier	$[0, tc/4]$	0	$1 - sin(\omega t)$
2 tilting	$[tc/4, tc/2]$	$4\,t/tc\ -\ 1$	0
2 raising barrier	$[tc/2, 3\,tc/4]$	1	$1 - sin(\omega(t - tc/4))$
4 untilting	$[3\,tc/4, tc]$	$4\,(1 - t/tc)$	1

with $\omega\ =\ 2\,\pi/tc$

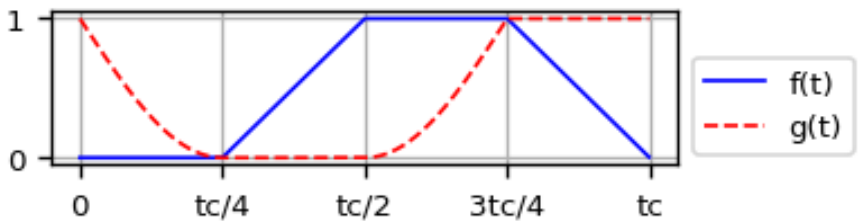

Figure 3 – Control functions f(t) and g(t)

The landscape at phase limits is given in figure A1 of the reference paper, and the evolution of the variables in figures A2.

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4. Single well memory

Using the same equipment employed to create the bistable memory — a colloidal particle in a fluid — we can design a single well memory in which the information entropy is controlled and stable, without the need for a potential barrier.
The reset to zero is obtained by getting the single well, either left or right, up to level U = 0, then by recreating the left well.
The application of the algorithms of the previous section are straightforward.
The result show that the work necessary for this reset to zero protocol is about $0.05\ k_B T$ for a cycle time of 900 s. The convergence is very slow, and the result can vary from 0.04 to 0.07 (0.04 on figure 4).

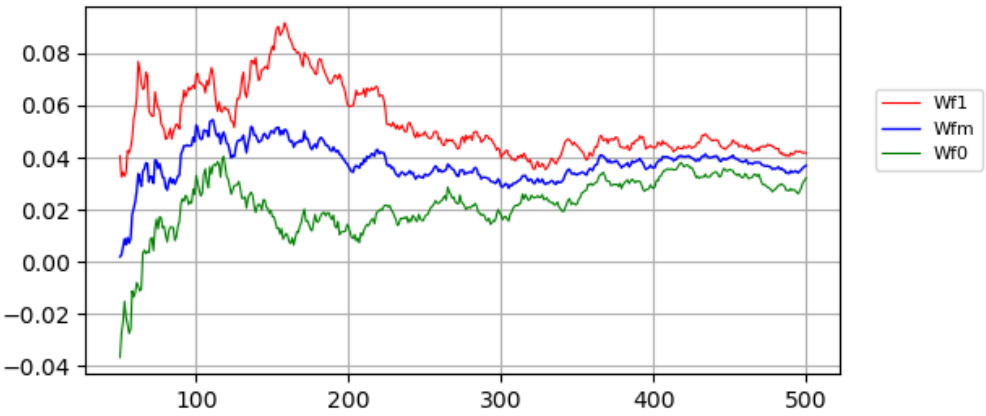

Figure 4 – Average work necessary for a reset to zero operation as a function of number of runs. $Wf0$ (resp. $Wf1$) for initial bit at 0 (resp. 1). And $Wfm = (Wf0 + Wf1)/2$.

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Bibliography

[1] J. Argouarc'h, « Limits of Landauer's principle and of the second law of thermodynamics », *Entropy: Thermodynamics – Energy – Environment – Economy*, vol. 7, nº Issue 1, 2026, doi: 10.21494/ISTE.OP.2026.1414.
[2] J. Argouarc'h, « Limites du principe de Landauer et du second principe de la thermodynamique », *Entropy: Thermodynamics – Energy – Environment – Economy*, vol. 7, nº 1, 2026, doi: 10.21494/ISTE.OP.2026.1413.
[3] Y. Jun, M. Gavrilov, et J. Bechhoefer, « High-precision test of Landauer's principle in a feedback trap », *Phys. Rev. Lett.*, vol. 113, nº 19, p. 190601, 2014.

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